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Thin cloud detection over land using background surface reflectance based on the BRDF model applied to Geostationary Ocean Color Imager (GOCI) satellite data sets



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ABSTRACT

Geostationary Ocean Color Imager (GOCI) sensor onboard the COMS (Communication, Ocean and Meteorological Satellite) launched in 2010 was primarily designed to provide high-frequency observations in and around the Korean Peninsula to ensure the thorough monitoring of ocean properties. Owing to its pixel resolution of 500 m and large set of spectral solar channels, GOCI can also be considered for applications related to the characterization of vegetation and the retrieval of aerosol properties over land. However, to apply it for the full characterization of land, it is mandatory to properly remove clouds from the images. Such a procedure has limitations when there is a lack of thermal bands, as is the case with GOCI. However, GOCI data are impacted by shadows and radiation scattering effects during the daily course of the sun. Although this yields strong directional effects, the bidirectional reflectance distribution function (BRDF) can be determined to a high level of accuracy. This information is used as a reference to detect clouds over land because surface BRDF varies slowly with time compared to that of clouds. The proposed algorithm relies on knowledge of the BRDF field derived from the application of a semi-empirical model that simulates the minimum difference between top and bottom of atmosphere reflectance values as the baseline of clear atmosphere. This step also serves to estimate background surface reflectance underneath clouds. Accuracy assessment of the new GOCI cloud mask product is appraised through a comparison with high-resolution vertical profiles of lidar data from the polar orbiting Cloud-Aerosol Lidar and Infrared Pathfinder Satellite Observation (CALIPSO). The results for the Probability Of Detection (POD) of all cloud types was found to be 0.831 for GOCI; this is comparable to that of MODIS (0.772). For the case of only thin cirrus, GOCI POD value was assessed to be 0.849, similar to that of MODIS, underlining the improved efficiency of determining thin cloud pixels.

1. Introduction

Clouds are considered as one of the most challenging natural targets to detect, especially over land (Kim et al., 2017; Liu and Liu, 2013; Sinyuk et al., 2007; Zhu and Helmer, 2018). On the other hand, determining the inherent optical properties for land and the atmosphere from remote sensing observations clearly requires only cloud-free scenes (Mei et al., 2017; Wu et al., 2018). Remotely sensed optical observations can identify thick clouds and determine properties such as reflectance, temperature, and type. In cases of thin clouds, however, close light interactions exist between the cloud and the land surface layers (Cahalan et al., 2001; Gao and Kaufman, 1995; Gao et al., 1998; Gao et al., 2002) and these interactions need to be taken into account. As thin clouds are semitransparent, the top of atmosphere (TOA) reflectance measured by satellites inevitably includes mixed cloud and surface radiation components.

Cloud mask algorithms generally rely on the use of some threshold values for the optical properties of clouds which serve to separate surface and cloud entities (Hagolle et al., 2010; Lotz et al., 2009; Mei et al., 2017). For instance, the Advanced Very High Resolution Radiometer (AVHRR) mainly uses two visible and three IR bands by determining optimal threshold values of channel histograms for cloud masking (Gesell, 1989; Saunders and Kriebel, 1988). Luo et al. (2008) screened cloudy areas in MODIS images at a scale of 250 m using the spectral features of the four land surface bands based on the threshold approach. Lotz et al. (2009) showed that the brightness of cloud

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reflectance is useful in classifying cloud, sun glint, and surface properties by determining threshold values from cloud top reflectance. As previous research initiatives have pointed out, however, the segregation of thin clouds still remains an issue.

In addition to the conventional spectral criteria, the temporal and spatial variations of a scene yield additional elements for cloud detection because the properties of land surfaces vary slowly over time and may show spatial contrasts, as opposed to those of clouds. In other words, cloud optical properties are more variable over short periods of time than the reflectance or emission of the underlying surface (Jedlovec et al., 2008; Lyapustin et al., 2008; Mei et al., 2017) and more spatially inhomogeneous than most surface natural targets (Martins et al., 2002). As an early approach of cloud masking, the International Satellite Cloud Climatology Project (ISCCP) cloud masking algorithm with two major components of visible and IR radiance was used to check the temporal variability of clouds (Rossow and Garder, 1993). Composite background reflectance data based on accumulated clear sky imagery was compared with current observations with potential cloud contamination. The same considerations to discard cloudy pixels were applied to MODIS data (Lyapustin et al., 2008). The quality of the results was comparable to that of the MODIS Collection 5 cloud products (MOD 35) over land surfaces. However, the variations of the bidirectional reflectance distribution function (BRDF) effects according to the spatial resolution were found to cause large uncertainties in the time series with regard to cloud screening (Lyapustin et al., 2008).

Regarding criteria based on spatial variations, Martins et al. (2002) developed a cloud masking method using the spatial variability of TOA reflectance in the visible bands of MODIS. The high spatial variability of clouds was also an effective criterion of cloud mask. However, this method was limited only to ocean areas, which do not necessitate a high spatial resolution of the final products. Thus, improving cloud detection is still a matter of great concern, not only to improve the accuracy of the assessment but also to secure the original spatial resolution of satellites.

Satellites equipped with solar spectral bands to monitor oceanic phenomena are used to measure the fluorescence, chlorophyll, and suspended sediments. This category of ocean color satellites, including the Sea-viewing Wide Field-of-view Sensor (SeaWiFS), Coastal Zone Color Scanner (CZCS), Medium Resolution Imaging Spectrometer on the Environmental Satellite (MERIS), and Sentinel-3 Ocean and Land Color Instrument (OLCI) have great potential to monitor vegetation cycles (Gobron et al., 2001; Guay et al., 2014; Wang et al., 2001) and retrieve aerosol properties accurately (Mei et al., 2017; Sayer et al., 2012; von Hoyningen-Huene et al., 2003, 2011). Ocean color missions enable the determination of spectral aerosol optical thickness bands (0.412–0.670 μm) and, additionally, NIR bands, which may be effective in interpreting surface properties (von Hoyningen-Huene et al., 2011). In particular, the retrieval of aerosol optical thickness (AOT) over land areas requires shortwave blue and visible spectral regions to subtract aerosol reflectance from measured TOA reflectance.

The geostationary ocean color imager (GOCI) aboard the Communication, Ocean, and Meteorological Satellite (COMS), which is the world's first geostationary ocean color sensor, observes spectra of oceans surrounding the Korean Peninsula with 6 visible spectral bands and 2 NIR bands (Faure et al., 2008; Ryu et al., 2012). In terms of marine remote sensing, it is quite meaningful to observe the ocean with higher temporal resolutions than existing polar orbiting ocean color sensors because the physical and optical characteristics of oceans are more variable than those of the land surface (Choi et al., 2012; Min et al., 2017; Shin et al., 2017; Volpe et al., 2011). In addition, the high temporal resolution of GOCI, with visible spectral wavelengths, is also favorable for applications in the domains of vegetation and the atmosphere. However, the poor performance of GOCI cloud mask mostly hampers any application involving further interpretation of the satellite signal related to vegetation and aerosol optical properties over land. The limitations of GOCI design, with respect to cloud detection, are the lack of IR spectral bands or additional shortwave (SW) IR spectral bands.

The observation capability of a geostationary satellite like GOCI differs from that of a typical polar orbiting satellite because the BRDF is well characterized due to changes in the solar position from sunrise to sunset (Geiger et al., 2008). Thus, with the changing position of the sun at every GOCI observation time, different visions of the optical properties of the surface-atmosphere can be inferred. The fine characterization of the BRDF offers a means to separate cloud slabs from the surface layer because the proportions of the land surface and atmosphere BRDF change with space and time. Therefore, in the case of GOCI, cloud detection and the derived cloud mask product could benefit from the BRDF data that reflects the inherent changes of surface properties.

In this study, thin cloud detection over land is improved using background surface reflectance based on a semi-empirical BRDF model applied to GOCI data. A previous study (Kim et al., 2017) suggested a method for assessing a cloud mask product over a limited area in the Korean Peninsula by simulating the background surface reflectance under cloud conditions using a BRDF model. First, areas with the highest NDVI values were selected to represent clear sky scenarios in which to distinguish thin clouds. As reported in a previous study that considered a GOCI full-disk image, the pixel value of one portion may not be sufficient to represent the entire image. Therefore, the baseline criteria for disentangling the atmosphere and surface compounds in the TOA reflectance of GOCI should be determined by considering not only the abnormal increases identified in reflectances that are inherent to the effects of clouds, but also the capability to extend the result to the full disk area. In this context, the overall objective of this study is to present an improved thin cloud detection algorithm applicable to all pixels of GOCI image. The new cloud mask algorithm was developed with the time series of background surface reflectance based on a BRDF model that served as a reference baseline for the clear atmosphere.

2. Study area and materials

2.1. GOCI imagery

GOCI sensor covers 2500 km × 2500 km of Northeast Asia, including South Korea, North Korea, Japan, and Northeast parts of China, as shown in Fig. 1. GOCI has 6 visible bands and 2 NIR bands with a spatial resolution of 500 m. The shortwave spectral bands of GOCI are mainly for ocean applications, such as yellow substance, turbidity, and chlorophyll. The specifications of GOCI are summarized in Table 1 (Cho et al., 2010). GOCI achieves daytime observations from 9:00 to 16:00 local time every hour. This high temporal observations enable the monitoring of the variations in the physical and optical properties of the ocean at an hourly frequency. Owing to these spectral and temporal characteristics, GOCI has high potential for applications to other fields.

2.2. CALIPSO and MODIS cloud products

To support the validation of GOCI cloud mask products, we used data from CALIPSO lidar level 1B (version 4.10) as benchmark validation data and MODIS-*Aqua* measurement (collection 6.1) as comparison data. CALIPSO is a space-based platform composed of three independent nadir-viewing measurements: Cloud-Aerosol Lidar with Orthogonal Polarization (CALIOP), Wide Field Camera (WFC), and Imaging Infrared Radiometer (IIR) (Winker et al., 2003). Among the three payloads, we selected Vertical Feature Mask (VFM) data derived from CALIOP to obtain cloud information along a vertical column. In addition, the accuracy of the new cloud mask derived from GOCI was first compared with those from CALIPSO and then with those from MODIS (Ackerman and Frey, 2015). Detailed specifications about these data sets are given in Table 2. To simultaneously compare various data sets, we attempted to select satellite images for similar time zones

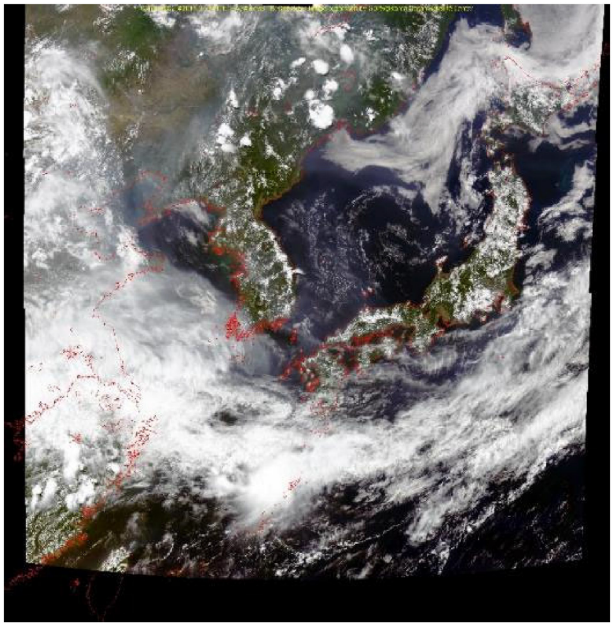


Fig. 1. Full disk sample of GOCI red-green-blue (RGB) images acquired at 04:00 UTC on 1 July 2014. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 1
Detailed Sensor characteristics of GOCI.

Band	Centre wavelengths	Band width	SNR[a]	Type
Band 1	0.412 μm	0.02 μm	1000	Visible
Band 2	0.443 μm	0.02 μm	1090	
Band 3	0.490 μm	0.02 μm	1170	
Band 4	0.555 μm	0.02 μm	1070	
Band 5	0.660 μm	0.02 μm	1010	
Band 6	0.680 μm	0.01 μm	870	
Band 7	0.745 μm	0.02 μm	860	NIR
Band 8	0.865 μm	0.04 μm	750	

[a] Signal-to-noise rate.

(Table 2).

CALIPSO, which follows *Aqua* with the general formation-flying concept, flies as an integral part of the Afternoon constellation (A-train and C-train) for Earth observations (Stephens et al., 2002). The satellite constellation aims at providing various types of scientific information for the same Earth scene almost concomitantly. The almost identical orbital characteristics of these satellites within a short time period allow intercomparison of their data. In fact, CALIPSO is no longer a member of the A-train but became a member of the C-train from September 13, 2018 because of a potential problem for maneuvering the capability of CloudSat. The C-Train follows a slightly different ground track because the C-Train orbits the Earth 16.5 km below the A-Train. In this study, fortunately, data derived from all measurements including GOCI, MODIS, and CALIOP covered the period from 2014 to 2015, which means CALIPSO had joined in orbit following *Aqua* in the experimental period.

3. Methods

3.1. Semi-empirical BRDF modeling for background surface reflectance

Before simulating background surface reflectance using the BRDF model, atmospheric correction was performed for multispectral bands of GOCI to estimate surface reflectance (SFC) using a previously calculated look-up-table (LUT) from the 6S atmospheric radiative transfer model (Nunes et al., 2008; Vermote et al., 1997; Wang, 2003). The input parameters of atmospheric constituents mostly used MODIS atmospheric products (MOD04, MOD05, and MOD07) due to their degree of reliability (https://modis-images.gsfc.nasa.gov/products.html). In cases where the MODIS atmospheric products were lacking, COMS MI products from the Korea Meteorological Administration (KMA) were used as a back-up. Although COMS MI atmospheric products covered the full GOCI hourly observation time, they were less accurate than MODIS atmospheric variables due to the limited number of channels for COMS MI (Kim et al., 2008). Because of this limitation, MODIS was preferred although a low daily variation of derived atmospheric constituents had to be assumed (Green et al., 2009) to serve as input parameters for GOCI atmosphere correction. Moreover, when we estimated the clear-sky surface reflectance prior to the development of the cloud mask for GOCI, the conventional threshold method of cloud masking was applied (see Section 3.2).

Once the surface reflectance of GOCI spectral bands was estimated, a semi-empirical BRDF model was used to simulate background surface reflectance. It supported two methods for thin cloud detection using GOCI. In the first, the BRDF model was utilized to simulate background surface reflectance for previously obtained time series imagery to determine the reflectance contribution of clear sky atmosphere only. In the second, the BRDF model was applied to simulate surface reflectance below existing current clouds and the results were compared with TOA reflectance to determine abnormal reflectance increase due to clouds. We adopted the semi-empirical BRDF model with Ross-thick and Li-sparse reciprocal (RTLSR) kernels to estimate the kernel coefficients and background surface reflectance (Roujean et al., 1992; Schaaf et al., 2002; Wanner et al., 1995). The theoretical basis of semi-empirical model contains three kernels with geometric, volumetric, and isotropic scattering reflectance, as follows:

$$\rho(\theta_s,\theta_v,\varnothing) = f_{iso} + f_{geo}\,k_{geo}(\theta_s,\theta_v,\varnothing) + f_{vol}\,k_{vol}(\theta_s,\theta_v,\varnothing) \quad (1)$$

where $\theta_s$is solar zenith angle, θ_v is viewing zenith angle, and $\varnothing$ is the relative azimuth angle between the satellite and solar azimuth geometry. The parameter f_{iso} is an isotropic coefficient of the Lambertian reflectance at nadir view ($\theta_v = 0$). The parameter f_{geo} is a coefficient of the LiSpare-Reciprocal geometric kernel k_{geo}, and the parameter f_{vol} is a coefficient of the Ross-Thick volumetric kernel k_{vol} (Roujean et al., 1992).

$$k_{vol} = \frac{4}{3\pi}\frac{1}{\cos\theta_s + \cos\theta_v}\cdot\left[\left(\frac{\pi}{2} - \xi\right)\cos\xi + \sin\xi\right] - \frac{1}{3} \quad (2)$$

where

Table 2
Detailed characteristics of the COMS GOCI, MODIS Aqua, and CALIPSO satellites.

	Orbit type (Altitude)	Local time in ascending node	Spatial resolution	Variable	Comparable imaging time
GOCI	Geo-synchronous (35,768 km)	–	500 m	New cloud mask	04:15 UTC
MODIS	Sun-synchronous (705 km)	Around 1:30 PM	1000 m	Cloud mask from MYD 35	04:25–04:55 UTC
CALIPSO	Sun-synchronous (705 km)	Around 1:30 PM	333 m	Vertical feature mask (VFM)	04:00–04:35 UTC

$$\cos\xi = \cos\theta_s \cos\theta_v + sin\,\theta_s sin\,\theta_v \cos\phi \quad (3)$$

The Li-Sparse geometric kernel is written as (Lucht et al., 2000):

$$k_{geo} = O - sec\,\theta_s' - sec\,\theta_v' + \frac{1}{2}(1 + cos\,\xi')\,sec\,\theta_v' \quad (4)$$

$$O = \frac{1}{\pi}(t - \sin t \cos t)(sec\,\theta_s' + sec\,\theta_v')$$

$$\cos t = \frac{h}{b}\frac{\sqrt{D^2 + (\tan\theta_s' \tan\theta_v' \sin\phi)^2}}{(\sec\theta_s' + \sec\theta_v')}$$

$$D = \sqrt{\tan^2\theta'_s + \tan^2\theta'_v - 2\tan\theta'_s}\,\tan\theta'_v \cos\phi$$

$$\cos\xi = \cos\theta'_s \cos\theta'_v + \sin\theta'_s \sin\theta'_v \cos\phi$$

$$\theta_s' = \tan^{-1}\left(\frac{b}{r}\tan\theta_s\right)$$

$$\theta_v' = \tan^{-1}\left(\frac{b}{r}\tan\theta_v\right),$$

The kernel coefficients of the semi-empirical BRDF model were determined independently for each spectral band and for each image pixel by inverting Eq. (1) with available cloud free observations during a 16-day composite period (Lucht et al., 2000; Schaaf et al., 2002; Yeom and Kim, 2015). To estimate background surface reflectance (ρ_{model}) using the BRDF model, at least seven observations of cloud free surface reflectance are required (Lucht et al., 2000; Schaaf et al., 2002). Therefore, the kernel coefficients of f_{iso}, f_{vol}, and f_{geo} were fitted with multiple linear regression only when at least seven observations of clear surface reflectance were secured.

From the set of determined BRDF model kernel coefficients, we simulated the modeled background surface reflectance (ρ_{model}) with the appropriate θ_s, θ_v, and $\varnothing$ values corresponding to GOCI geometry. The viewing zenith angle and azimuth angle are fixed for each GOCI observation point. In contrast, the solar zenith angle and azimuth angle are calculated for each GOCI pixel as functions of a specific time, latitude, and longitude. Fig. 2 shows the main conceptual diagram for the simulation of the desired GOCI observation time.

3.2. Algorithm for cloud masking

The main concept of our new cloud mask algorithm is based on the temporal variation of the difference in reflectance between TOA and background surface reflectance from the BRDF model to distinguish clear and cloudy skies. If a GOCI pixel is deemed clear, the atmospheric reflectance and land surface reflectance appear without cloud effects. Therefore, the clearer the atmosphere, the lower the difference between the TOA and surface reflectance. This implies that the minimum difference in reflectance between the TOA and land surface represents the clearest day during a certain period of time. In the presence of clouds, TOA reflectance appears to be higher than the clear atmosphere reflectance, indicating a difference in reflectance beyond the clearest atmospheric effects. Finally, the difference between the TOA and background surface reflectance was used as the main criterion to appraise the quality of the thin cloud mask in this study. Therefore, the minimum reflectance difference before the target date was estimated as a baseline of the clearest day of atmospheric reflectance, and the reference baseline was compared with the current reflectance difference in the TOA and surface. Subsequently, cloud effects were confirmed if the current reflectance difference was higher than the previously determined baseline of the clearest atmospheric reflectance by a certain threshold value.

Fig. 3 shows a schematic diagram of the new cloud masking algorithm using GOCI data. The new GOCI cloud mask algorithm is mainly divided into two steps.

Firstly, the simple threshold method using each of single band and band ratio was used to discriminate “thick cloud” areas as a process in step 1. Such areas can be classified well through simple threshold methods because thick clouds have relatively higher reflectance and colder temperatures than any natural objects on the surface (Ackerman et al., 1998). In step 1, pixels that were not identified as “thick cloud” are subjected to step 2 to determine whether they are “thin cloud”, “probably clear”, or “certainly clear”. Step 2 is a newly suggested thin cloud method used in this study, in which the contribution of surface reflectance under cloud areas is reflected and the clearest atmosphere with the minimum reflectance difference between TOA and background surface reflectance for the previous 16 days is selected from the times series cases. In this study, we used mainly short wavelength regions of GOCI imagery, e.g., the blue band 1 (0.412 μm), to classify thin clouds because the effects of surface reflectance become lower as the wavelength becomes shorter in visible spectral properties of land. This means that the reduction of surface reflectance in the blue band makes it easier to distinguish thin clouds clearly, similar to the method of retrieving aerosol optical properties (von Hoyningen-Huene et al., 2011). Furthermore, the accuracy of the new thin cloud algorithm of GOCI shows less variation according to the wavelength due to the use of only the blue band. We defined the algorithm concept expressed as the following Eqs. (5)–(6);

$$DIFF_i = TOA_i - BRDF_i + |e_i|,\ i = -16, 0 \quad (5)$$

$$e_i = BRDF_i - SFC_i \quad (6)$$

where $DIFF_i$ is the difference between TOA_i reflectance and BRDF model-based surface reflectance ($BRDF_i$) at i day. i day represents the value from target day ($i = 0$) to 16 days before the current day ($i = -16$). e_i is BRDF modeling error between simulated surface reflectance ($BRDF_i$) and atmospherically corrected surface reflectance (SFC_i) to reflect uncertainty of the semi-empirical BRDF model (Han et al., 2004; Yeom and Kim, 2015). When determining the presence of thin clouds, we consider the error of the BRDF model as the margin

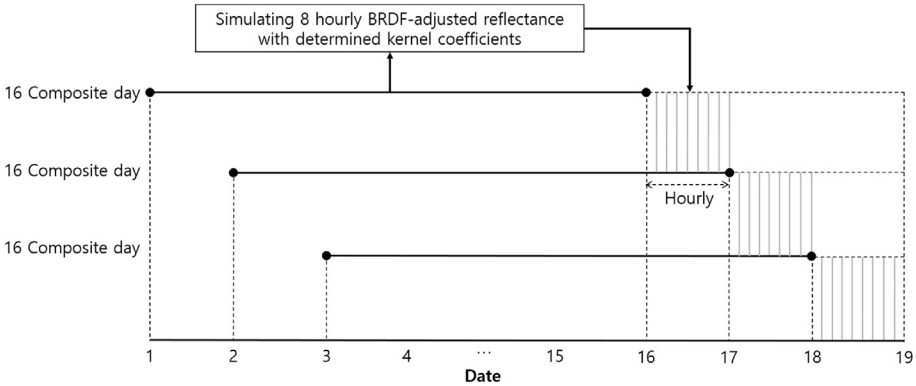


Fig. 2. Scheme of 16 day composite period for simulating 8 hourly BRDF-adjusted background surface reflectance with determined kernel coefficients.

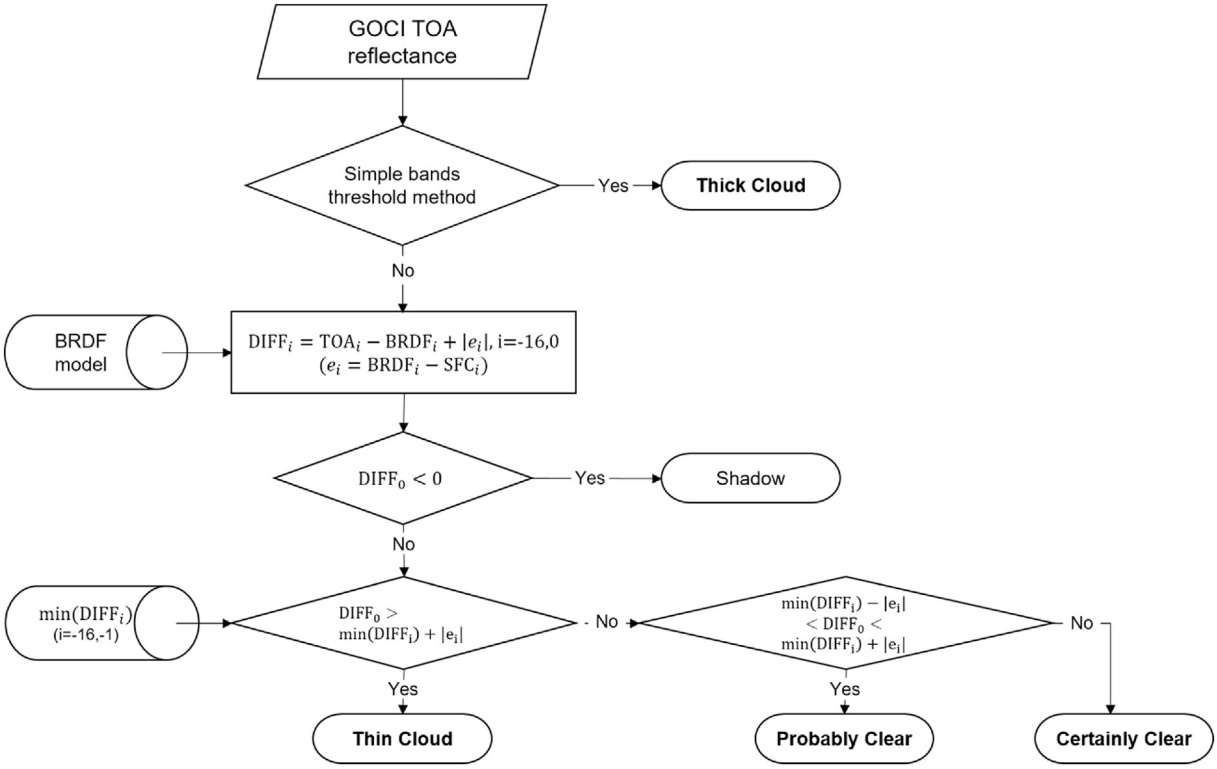


Fig. 3. Schematic diagram of the cloud masking algorithm using GOCI data.

value of the threshold (e_i) for more reliability of determining the thin cloud area. In addition, we calculated the minimum $DIFF_i$ (i = −16,-1) in the previous 16 days to find the atmospherically clearest day on the assumption that at least one day might be a clear day. The definition of minimum $DIFF_i$ (*MIN*) is as follows:

$$MIN = \min(DIFF_i),\ i = -16, -1 \tag{7}$$

For reference, the *MIN* values for every hour were estimated for the same time zone of GOCI data to minimize the BRDF effect according to the variation of the solar zenith angle.

In the flowchart of cloud mask, if the $DIFF_0$ value was negative, the pixel was assigned as a "shadow" area because the cloud shadow makes TOA reflectance observed by GOCI satellite abnormally lower than the simulated surface reflectance. Furthermore, for shadow detection, the longest wavelength of GOCI band 8 (0.845–0.885 μm) was used to classify the cloud shadow because the diffusive radiation in the shadows is relatively smaller at longer wavelengths (i.e., NIR and SWIR bands), making the pixels in the shadowed area darker than their surroundings (Luo et al., 2008; Zhu and Woodcock, 2012). In contrast, if the pixel had a positive value of $DIFF_0$, "thin cloud", "probably clear", and "certainly clear" were classified through comparison with the *MIN* value, which is the reference baseline of the clearest day before the target day. If $DIFF_0$ is greater than the sum of *MIN* and absolute model error (|e_i|), the pixel was assigned as "thin cloud" because we defined "thin cloud" as having values higher than the sum of the *MIN* and model error (|e_i|). This means that TOA reflectance is abnormally increased due to cloud occurrence. After "thin cloud" processing, residual pixels were defined as "probably clear" or "certainly clear". If the $DIFF_0$ value was smaller than *MIN* plus the model margin error (|e_i|) and greater than *MIN* minus the |e_i|, it was classified as "probably clear". Finally, if $DIFF_0$ was less than *MIN* minus |e_i|, the pixel was assigned "certainly clear" because the current condition of the atmosphere would be similar to the clearest atmospheric condition during the last 16 days.

3.3. Parallax correction for multi satellites for GOCI and MODIS

While comparing some satellite-derived scientific data, parallax error may arise between the two types of satellite data. As the satellite zenith angle and the altitude of clouds increase, location error of clouds due to the parallax error would be increased as well (Lee et al., 2011). Therefore, the parallax error should be eliminated to achieve accurate cloud masking results. In particular, the line of sight at the ground could be dissimilar depending on the orbital characteristics. Because we validated two types of satellite data with different orbital characteristics, the location error of clouds inevitably appeared. Thus, parallax correction was required to match the location of cloud ground footprints for various satellite observations (Pešice, 2009; Vicente et al., 2002; Wang et al., 2011).

Fig. 4 shows a schematic diagram of the parallax correction used in this study between COMS and CALIPSO. This parallax correction method has already been described by Kim et al. (2017), in which the method presented by Lee et al. (2011) was applied in a reversed manner. Unfortunately, GOCI cannot observe the altitude of clouds, and thus cloud footprints observed from CALIPSO were translocated to the point of looking from GOCI (Kim et al., 2017). In other words, through the parallax correction, cloud pixels of CALIPSO for point P_c (actual cloud position) could be moved to point P_o (modified cloud position) to compare with GOCI pixels without any parallax errors. Several physical values such as the altitude of clouds (M_{cld}), radius of the Earth (R_e), distance from the surface of the Earth to COMS (M_{sat}~36,000 km) on the geostationary orbit, and satellite zenith angle for point P_c (θ') are readily available. Applying a quadratic formula and the sine/cosine rules multiple times with these values, the modified satellite zenith angle (θ) for CALIPSO can be finally written as follows:

$$\therefore \theta = arcsin\left(\frac{R_{sat}R_{cld}}{R_eR_c}\,sin\varnothing\right), \tag{8}$$

where R_{sat} is the distance from the center of the Earth to COMS, R_{cld} denotes the distance between the cloud and the center of the Earth, and

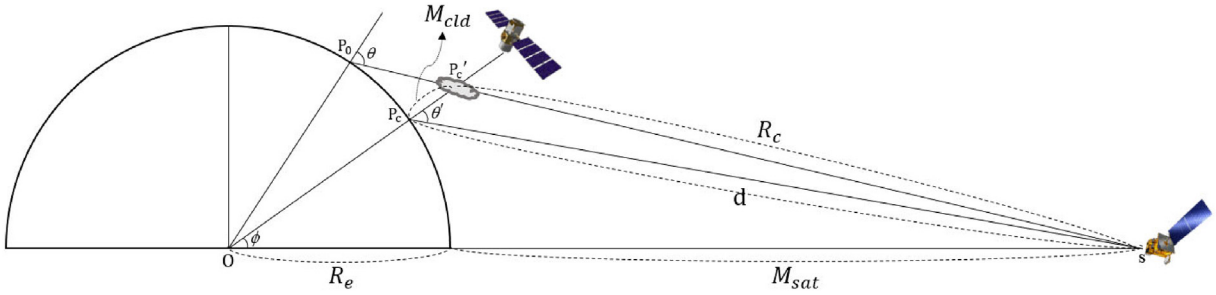


Fig. 4. Schematic diagram of parallax correction between COMS and CALIPSO satellites.

R_c represents the distance from the cloud to COMS. For detailed information, the following variables need to be acquired for θ:

$$d = \sqrt{(R_{sat} + R_e \sin\theta')(R_{sat} - R_e \sin\theta')} - R_e \cos\theta', \tag{9}$$

$$R_c = \sqrt{M_{cld}{}^2 + d^2 - 2dM_{cld}\cos\theta'}, \tag{10}$$

$$\varnothing = arcsin\left(\frac{d}{R_c}\sin\theta'\right) - arcsin\left(\frac{dR_{cld}}{R_c R_{csat}}\sin\theta'\right). \tag{11}$$

Acquiring these variables through Eqs. (9)–(11) and applying them to Eq. (8), the parallax correction can be successfully performed.

Moreover, to verify the cloud masking results using GOCI, MODIS data were used for comparison. The parallax error may occur as differences not only in the orbital characteristics, but also in the instrumental characteristics. Therefore, the parallax error between MODIS and CALIOP would obviously exist as well. We performed the parallax correction for these two satellites based on the method presented by Wang et al. (2011). They suggested a parallax correction method for two sun-synchronous satellites, *Aqua* and CloudSat, which were members of the A-Train constellation during the study period. Therefore, the approach could be directly adapted to this study given similar conditions of satellite characteristics. Thus, MODIS-CALIOP parallax correction was performed by applying CALIPSO data instead of CloudSat using the method of Wang et al. (2011). Detailed explanation for the parallax correction between MODIS and CALIOP is not provided in this paper.

4. Results

4.1. Visual inspection of cloud masking

In this study, the newly suggested cloud mask algorithm was analyzed by estimating cloud mask maps using GOCI imagery for the study area from 1 January 2014 to 30 June 2015. When selecting the examples of cloud mask from GOCI, normal meteorological phenomena of the study area such as Typhoons and the East Asian summer monsoon front (known as Changma in South Korea) were included with seasonal cases of cloud mask results. Fig. 5 shows example results of cloud mask maps from GOCI imagery. Figures in the left column from (a) to (f) are RGB combined images from GOCI satellite and the figures in the right column (g-l) correspond to results of the cloud mask. The cloud mask map was classified as “thick cloud” (dark blue), “thin cloud” (light blue), “probably clear” (light green), and “certainly clear” (light yellow) according to the flowchart of the cloud masking algorithm in Fig. 3. In Fig. 5, most cases well captured semitransparent thin cloud areas, which are represented as “light blue” when compared with the corresponding RGB images of GOCI. In addition, numerous “thin cloud” (light blue) areas appeared on the edge around the generally “thick cloud” (dark blue) area.

In the case of thin clouds, first, a light white area from East China appeared to spread widely across North China, as shown by the red circle in Fig. 5 (a) RGB image at 04 UTC, 28 July 2014. The corresponding GOCI cloud mask map in Fig. 5 (g) is well classified as “thin cloud” area when compared with the RGB image. Because of the semi-transparency of thin clouds, the spatial pattern of the land surface under the cloud area in the red circle of Fig. 5 (a) faintly appears unlike the thick clouds, indirectly indicating that thin cloud detection is difficult without considering the contribution of surface reflectance underneath clouds. In Fig. 5 (b-h) and (c-i), the cloud mask maps of GOCI are consistent with the locations of “thick cloud”, “thin cloud”, “probably clear”, and “certainly clear” areas despite the absence of GOCI IR channels. In particular, in the case of the red circle in Fig. 5 (d), the “thin cloud” area over North Korea through the Yellow Sea appears to be well captured in the results of cloud mask in Fig. 5 (j).

In the cases of summer monsoon (Fig. 5 (e)-(k)) and typhoon (Fig. 5 (f)-(l)), the spatial patterns of clouds according to the occurrence of meteorological phenomena could be well classified into cloudy or clear cases. In Fig. 5 (e) and (k), the long monsoon cloud front extended from east to west, which induced many clouds in Northeast Asia, in GOCI RGB image. “Thin cloud” areas were also found to be dispersed around “thick cloud” areas in the red circle in Fig. 5 (e) RGB image. Lastly, the spatial distribution of thin clouds attributable to the weak monsoon cloud front due to the northward Typhoon NAKRI is well represented in the results of the cloud mask in Fig. 5. (l) when compared with Fig. 5 (f). In conclusion, “thick cloud” is easy to detect with the step 1 process, but “thin cloud” is difficult to detect using the conventional threshold method due to limited GOCI spectral bands. Nevertheless, the suggested algorithm can effectively classify semitransparent thin clouds through step 2 by applying background surface reflectance from the BRDF model.

Previous studies using cloud mask algorithms with standard deviation methods (Choi et al., 2016; Martins et al., 2002) improved the cloud detection accuracy and showed reasonable results of the cloud mask. However, these methods inevitably degraded the original spatial resolution of satellite image. In this study, spatial variation of window pixels was not used to obtain pixel based maps of the cloud mask. Therefore, the original spatial resolution of 500 m of GOCI imagery was retained in the cloud mask results. In Fig. 5 from (g) to (l), complicated and variable spatial patterns of thin and thick clouds are well represented in the maps of GOCI cloud mask. We believe that this is one of the effective advantages of our cloud mask algorithm because the spatial resolution of the cloud mask directly affects the determination of the spatial resolution of applications such as to vegetation on land and aerosol optical properties (AOPs) in the atmosphere.

4.2. Quantitative evaluation of cloud masking

The results of the proposed thin cloud mask from GOCI were validated with CALIPSO satellite products, which were used as reference data. Lidar observations of CALIPSO support accurate detection, monitoring, and interpretation of cloud properties (Winker et al., 2007). In addition, MODIS Collection 6.1 cloud mask products (MYD35) from Aqua were also used for validation. According to Table 2, each of the three satellites has different orbit types and revisit times, meaning that

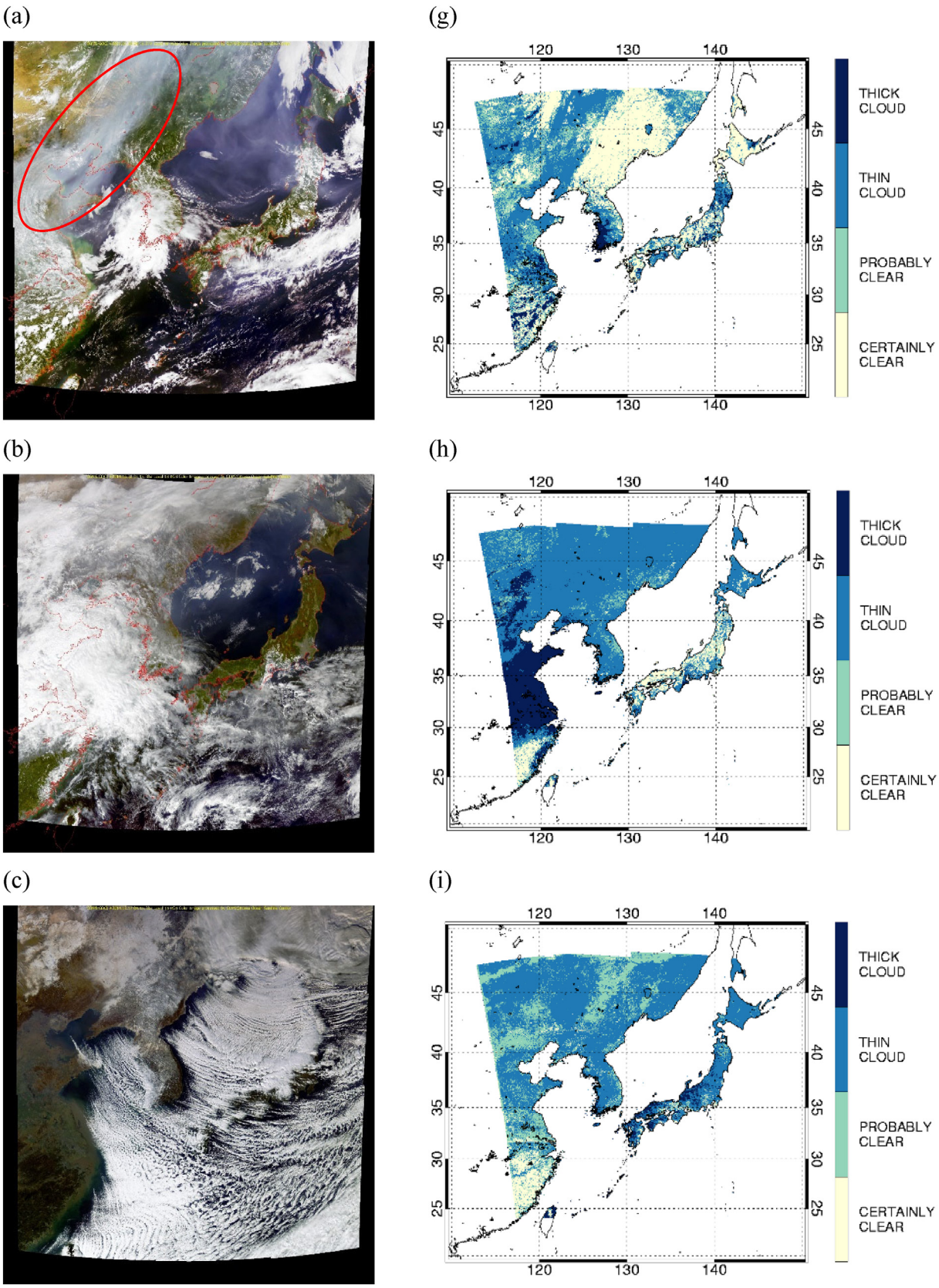


Fig. 5. RGB images (a-f) and cloud masking results (g-l) for GOCI in (a, g) summer (2014-07-28 04 UTC), (b, h) autumn (2014-10-30 04 UTC), (c, i) winter (2014-12-17 04 UTC), (d, j) spring (2015-04-26 04 UTC), (e, k) a Changma case (2014-07-01 04 UTC), and (f, l) a Typhoon case (2014-08-02 04 UTC).

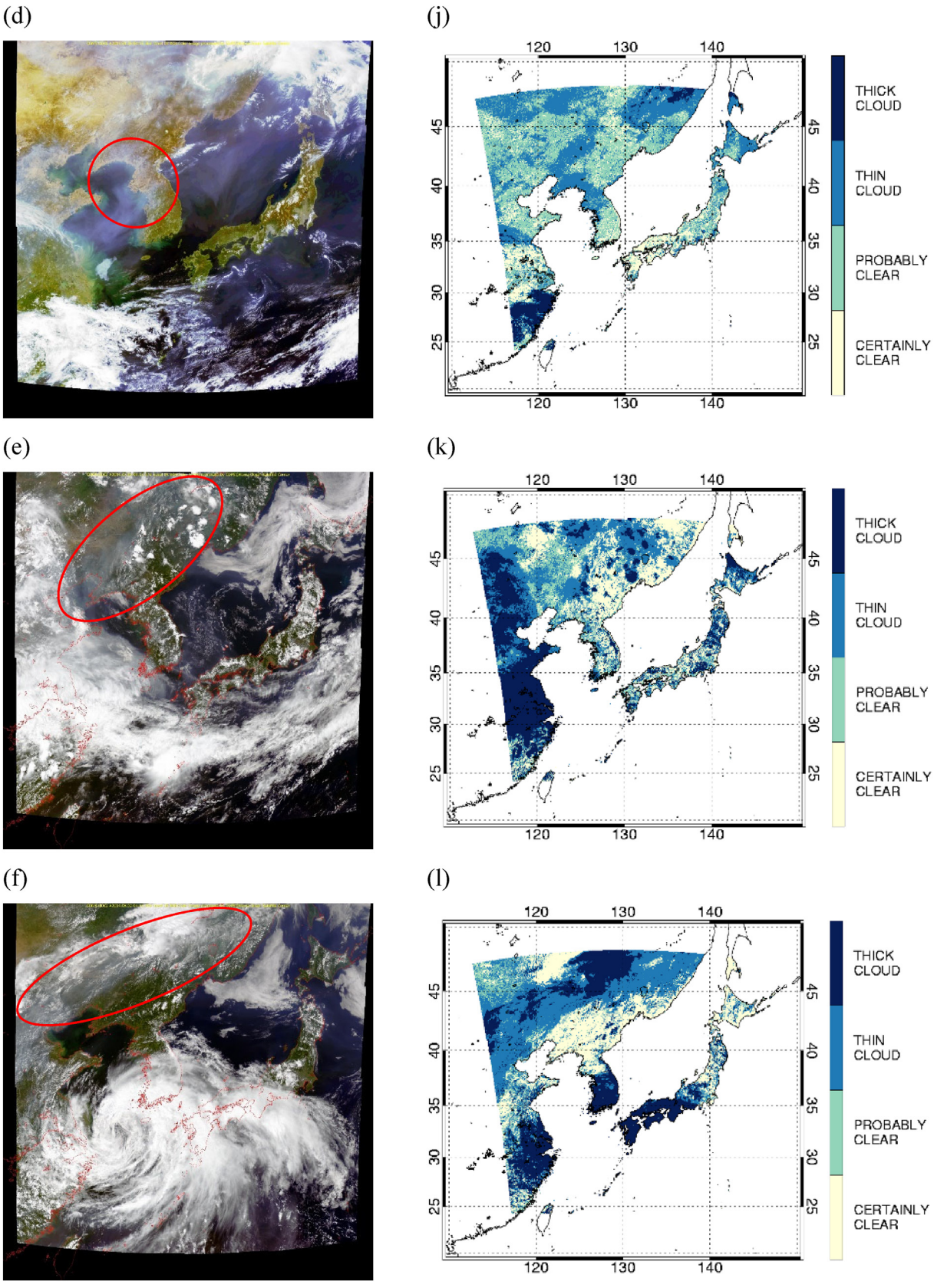


Fig. 5. (*continued*)

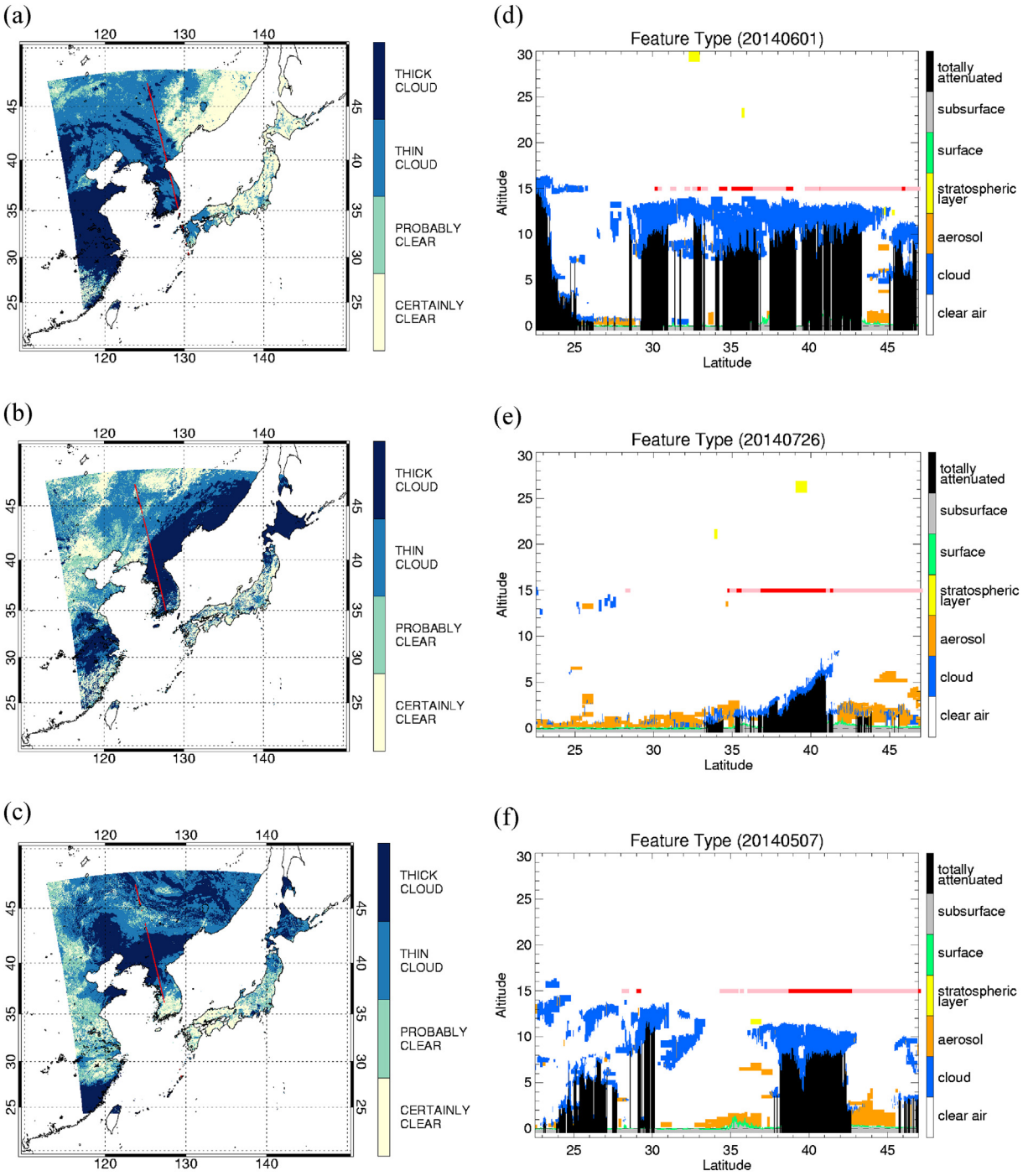


Fig. 6. CALIPSO satellite trajectory (red line) on a GOCI cloud mask map (a-c) and vertical profile for CALIPSO vertical feature mask (VFM) (d-f) on (a, d) 2014-06-01, and (b, e) 2014-07-26, and (c, f) 2014-05-07. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

spatial and temporal collocation of each cloud product from distinctive satellites should be performed to secure the accuracy of validation results for GOCI cloud mask. In the case of geometry discrepancy, we performed parallax correction, as described in Section 3.3 to collocate geometrically different observation orbit types of satellites. Observation projections of GOCI and MODIS were resampled to CALIPSO projection to reduce parallax errors of cloud mask results. In the case of temporal discrepancy of each satellite, we simply selected the nearest observation time of GOCI and MODIS to the CALIPSO observation time. Because geostationary GOCI observes the study area every hour, the time difference from CALIPSO is 30 min at most (worst condition) and 15 min at least (best condition). GOCI sequentially observes each part of the full disk area with the two-dimensional staring frame capture method (Cho et al., 2010) and covers the full disk area in approximately 15 min. Finally, we assumed that the variation of cloud is insignificant within the observation time of GOCI when validating with CALIPSO cloud products. MODIS and CALIPSO make observations with time zones similar to that of the A-train and similar local time in the ascending mode according to Table 2.

Fig. 6 shows examples of temporally and spatially collocated data sets for GOCI and CALIPSO during the study periods for comparison with each cloud mask product. In Fig. 6, left column images from (a) to

(c) are trajectories (red line) of CALIPSO on the classified cloud mask maps of GOCI, and right column images represent the corresponding Vertical Feature Mask (VFM) of CALIPSO. In the case of VFM images in Fig. 6 (d-f), the vertical legends on the right are atmospheric profiles categorized by CALIPSO lidar sensor, and the vertical profiles of clouds can be showed as the thickness of the blue color area. Thick clouds are mostly marked as black color area under cloud in VFM because incoming solar radiation would be totally attenuated by thick clouds. In addition, the horizontal line at 15 km in the VFM image corresponds to the thin cloud (light red line) and thick cloud (red line) areas located on the CALIPSO trajectory on the left image of GOCI cloud mask results. In Fig. 6 (a) and (d), the spatial distributions of thin cloud (light red line) and thick cloud (red line) areas from GOCI cloud mask well match with cloud patterns of CALIPSO VFM. It spatially coincided with the types of clouds that appear in the CALIPSO trajectory passing through the southern part of Japan, the Korean Peninsula and northern part of China. Furthermore, when comparing each GOCI cloud map and CALIPSO VFM in Fig. 6 (b), (e), and (c), (f), the thin cloud and thick cloud areas are spatially well matched over the land surface area despite the absence of IR channels in GOCI, similar to the previous figures of cloud mask. In addition, in Fig. 6 (f), GOCI cloud mask results show a thin cloud area 35° latitude in South Korea and at 44° latitude in the north part of China, but CALIPSO VFM classified these areas as aerosol. Our algorithm is mainly intended for classifying semitransparent thin clouds as it can detect abnormal increases of atmospheric reflectance through comparison with stable background surface reflectance. Nevertheless, as shown in Fig. 6 (f), this can be applied to thick aerosols as thick aerosols also increase with atmospheric reflectance for broad visible spectral wavelength similar to cloud optical properties. Conversely, another assumption might be that the VFM misclassified thin clouds as aerosol. VFM is known to occasionally make false detections through its horizontal averaging procedure and may also misclassify noise or aerosol as cloud and vice versa (Holz et al., 2008; Marchand et al., 2008). Therefore, further research is needed to address this issue since the discrimination of thin cloud and aerosol is also very challenging (Lee et al., 2013).

Finally, we statistically validated the cloud mask results of GOCI from 1 January 2014 to 30 June 2015 over the Northeast Asia region, using spatially and temporally collocated matchup data sets. Table 3 shows the validation results of GOCI cloud detection compared to the CALIPSO VFM images. As mentioned earlier, the official MODIS cloud mask product (MYD35) from NASA was also compared with CALIPSO data for comparative analysis because the accuracy of GOCI cloud mask results can be indirectly evaluated through the results of MODIS validation with CALIPSO within the same region and time of GOCI observation. When performing validations of GOCI and MODIS with CALIPSO, we separated the validation cases according to “all cloud” and “thin cloud only” cases because the results of detection of thin cloud should be separately presented, which is the main purpose of this algorithm. In addition, the cloud types of CALIPSO were classified as overcast, stratocumulus, cumulus, altocumulus, altostratus, cirrus, and deep convective. Among various cloud types, altostratus and cirrus

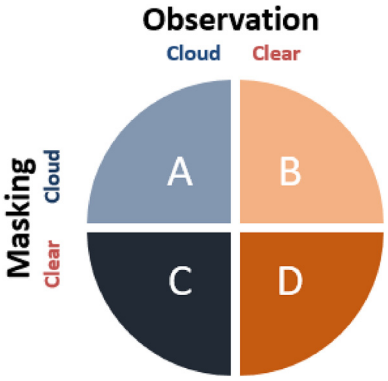


Fig. 7. Schematic diagram of the contingency table.

were assigned as thin clouds and the remainder was classified as thick clouds.

We calculated the POD for each case as quantitative validation results. Fig. 7 shows the diagram of the contingency table for two cloud masking results. “Observation” means CALIPSO VFM; “Masking” could be GOCI cloud mask result when we tried to acquire the validation result for GOCI cloud masking. The validation results including POD in Table 3 can be obtained as follows with Fig. 7:

$$POD = A/(A + C) \quad (12)$$

$$FAR = B/(A + B) \quad (13)$$

$$MAR = C/(A + C) \quad (14)$$

Experiment 1 and 2 in Table 3 were performed for all cloud cases including thin cloud cases from GOCI and MODIS respectively, and experiment 3 and 4 were executed only for thin cloud (altostratus, cirrus) cases of CALIPSO using GOCI and MODIS data. In the all cloud case of GOCI for experiment 1, the probability of detection (POD) value from step 1 was remarkably improved by applying the newly suggested step 2. Through step 2, residual semitransparent clouds could be effectively detected despite the limitation of GOCI spectral bands. Compared to the POD of MODIS cloud products (0.772), GOCI has higher POD value at 0.831 for the all cloud cases. Some previous studies have also validated MODIS cloud mask products using CALIPSO (Hagihara et al., 2010; Holz et al., 2008). Wang et al. (2016) validated a MODIS cloud mask with CloudSat-CALIPSO cloud profiles and reported an agreement of 77.8% between MODIS and CALIPSO, indicating that estimated POD values of MODIS within GOCI observation area were similar despite of differences in time and location. In the case of false alarm rate (FAR), GOCI FAR with 0.034 was overestimated but missing alarm rate (MAR) of GOCI with 0.169 was lower than the MODIS MAR of 0.228. The higher FAR and lower MAR values of GOCI indicate that our cloud mask algorithm is slightly more restricted than MODIS cloud mask criteria. However, the difference of FAR and MAR for GOCI and MODIS are insignificant for all cloud cases, indicating that the suggested method is effective for detecting clouds.

Table 3
Validation results of the cloud detection method of GOCI compared to CALIPSO data. Comparative analysis was also performed using MODIS and CALIPSO data.

	Experiment 1	Experiment 2	Experiment 3	Experiment 4
Sensor	GOCI	MODIS	GOCI	MODIS
Cloud types of CALIPSO	All cloud	All cloud	Thin cloud (altostratus, cirrus)	Thin cloud (altostratus, cirrus)
POD (only Step1)[a]	0.638	–	0.338	–
POD (Step1 + Step2)[b]	0.831	0.772	0.849	0.849
FAR	0.034	0.019	–	–
MAR	0.169	0.228	0.151	0.091
Sample number	200,869	140,425	44,296	24,996

[a] Only step 1 was applied for calculating the POD of GOCI.
[b] Step 1 and step 2 were applied for calculating the POD of GOCI for experiments 1 and 3.

For “thin cloud only” validations of GOCI in experiment 3, the accuracy of thin cloud detection of GOCI was significantly improved by applying step 2 from 0.338 until 0.849, as shown in Table 3, meaning that the newly suggested method is effective at detecting thin clouds despite the limitation of GOCI spectral bands. In the thin cloud case, POD of MODIS was 0.849, which is similar with that of GOCI. In addition, MAR showed only small difference between GOCI (0.151) and MODIS (0.091), indicating that the suggested cloud mask based on the simulated background surface reflectance is effective at detecting thin clouds with low values of FAR and MAR.

5. Discussion

In this study, cloud products of both GOCI and MODIS were compared with those of CALIPSO to evaluate the performance of these sensors for cloud masking. Nevertheless, a direct comparison between GOCI and MODIS cloud masking is required because MODIS not only detects clouds using state-of-the-art sensors, but its algorithm has also been validated over a long time worldwide. However, when directly comparing the cloud results of GOCI and MODIS, the parallax error should also be considered for accurate validation of each cloud masking results due to their geometry discrepancy as mentioned earlier. In particular, the cloud altitude should be used for reliable parallax correction. However, MODIS spectral bands are limited in estimating cloud altitude when compared with the lidar sensor of CALIPSO. Therefore, we used the cloud altitude data of CALISPO for comparing GOCI with MODIS. In other words, in the CALIPSO observation trajectory, the cloud results of GOCI and MODIS were compared for pixels that are closest in time and location to the two satellites. Comparing MODIS products with GOCI cloud products, the result of POD is higher (0.855) than the value compared with CALIPSO for the all-cloud case in Table 3. The result is quite reasonable when we evaluate GOCI cloud mask with the POD only. However, FAR is also increased over 0.167 in comparison with the false alarm between GOCI and CALIPSO. This means GOCI cloud mask was overestimated as cloud area when compared with MODIS cloud products. This is similar to that of GOCI cloud mask results, which has higher POD and higher FAR than MODIS results over all cloud conditions, comparing GOCI and MODIS to CALIPSO (Table 3). Although direct comparison with MODIS is absolutely necessary, some constraints on validation exist. Even though we performed parallax correction, differences in viewing zenith and azimuth angle are expected because only CALIPSO match-up is used. Moreover, the imaging time zone of MODIS arrives later than that of the other two satellites. Consequently, the value of FAR increases because of the long time interval between GOCI and MODIS. Even though this examination will need further analysis using state-of-the-art cloud products or ground measurements, the pattern of GOCI cloud mask map is verified to be similar to cloud mask of MODIS.

Although the accuracy of the newly suggested method of GOCI cloud mask was found to be reliable according to statistical values, this algorithm involves some issues mainly due to limitations of GOCI spectral bands biased towards ocean color observation. The upper left corner of GOCI images in Fig. 5 shows the location of Horqin sand land, which is one of the areas of most serious concern for land desertification in China (Zhao et al., 2007). According to the cloud mask results shown in Fig. 5 from (g) to (l), the proposed algorithm misclassified these areas as cloud areas although they were under clear sky conditions. Because the bright reflectance of desert areas reduces the difference in reflectance between land surface underneath thin clouds, it becomes difficult to distinguish features with similarly high reflectance. This is one of the main disadvantages of the limited channels of GOCI. Therefore, additional approaches of GOCI thin cloud detection should be considered for bright land surface areas.

6. Summary and conclusions

This study was conducted to determine means for applying the visible spectral channels of ocean color sensors mounted on GOCI satellite to vegetation or AOPs, especially over the land surface. Although the main purpose of GOCI is to monitor ocean phenomena around the Korean Peninsula, the spectral wavelength regions of ocean color sensors are very attractive for land and atmosphere applications. However, clouds over the land surface present serious obstacles limiting its application because the land surface has higher reflectance than the ocean area. Therefore, a new cloud detection method is required to expand the application fields of GOCI from ocean to land vegetation or atmospheric AOPs over land surface. In this study, we suggested a new approach of GOCI thin cloud detection by simulating the background surface reflectance, which is a crucial factor of our method for discriminating thin cloud over the land surface. The semi-empirical BRDF model was mainly used to determine temporal background surface reflectance as a baseline of clear atmosphere and to estimate background surface reflectance underneath clouds. In particular, simulating land surface reflectance with high sensitivity to geometric characteristics is useful in accounting for the portion of upward surface reflectance from the mixed reflectance observed by GOCI with high temporal resolution. In conclusion, this new concept of cloud mask algorithm can effectively detect thin clouds over the land surface area even with the limited spectral bands of GOCI unlike previous cloud mask algorithms. We validated the results of the cloud mask from GOCI with CALIPSO data, which were taken as reference considering the high-resolution vertical profiles of the lidar measurement system of the satellite. POD values of the new GOCI cloud mask algorithm was 0.831 under the all-cloud condition, whereas MODIS POD was 0.772, indicating that the suggested cloud mask effectively detected thin and thick clouds over the study area. Compared with previous results of MODIS cloud masking using CALIPSO, the suggested new cloud mask method was considered to be useful for detecting thin clouds with GOCI products, especially over the land surface. Given that the MODIS POD and GOCI POD values were 0.849, our cloud mask algorithm exhibited similar performance to that of MODIS for thin cloud detection. Furthermore, the FAR and MAR of GOCI cloud mask results were also low similar to those of MODIS, indicating that not only cloud cases but also clear sky conditions could be well classified. Although the statistical values of GOCI thin cloud mask suggest reliability, misclassifications of cloud detection in bright desert areas are one of main issues of this algorithm and will require further improvement.

We believe that the new thin cloud masking algorithm has the potential for application to future satellites. GOCI-II, which is the next-generation payload of GOCI, is under development by the Korea Aerospace Research Institute (KARI) and is scheduled to be launched in 2020. The proposed cloud masking algorithm would also be suitable for GOCI-II because it has a similar spectral range as GOCI but with other advanced spectral bands. Furthermore, several ocean color satellites are also planned to be launched in the near future by NASA, such as the Geostationary Satellite of Coastal and Air Pollution Events (GEO-CAPE) and Plankton, Aerosol, Cloud and ocean Ecosystem (PACE), both of which will be geostationary. Therefore, we expect that our new cloud masking algorithm using the simulated background surface reflectance from BRDF model can not only be implemented in cloud detection but also extend the application fields of satellites such as GOCI-II, GEO-CAPE, and PACE from the ocean to land or atmosphere observations. In addition, our new method would provide an alternative means to improve the accuracy of cloud masks for existing EOS satellites, such as MODIS, Himawari-8, and Geostationary Korea Multi-Purpose Satellite (Geo-KOMPSAT)-2A Advanced Meteorological Imager (AMI).

Author contribution

Jong-Min Yeom: Concptualization, Methodology, Validation,

Visulization, Writing-Original Draft
Jean-Louis Roujean: Formal analysis, Writing-Review & Editing
Kyung-Soo Han: Writing-Review & Editing, Funding acquisition
Kyeong-Sang Lee: Formal analysis, Writing-Review & Editing
Hye-Won Kim: Methodology, Validation, Supervision, Funding acquisition, Writing-Review & Editing

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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